

## Homework 3

### Definitions

Define the following terms/concepts. (The language of probability - sample space, event, random trial, union, intersection, random variable, probability distribution and the Law of Large Numbers - was covered on **Homework 2**; review those definitions before starting.)

- a.) Population distribution
- b.) Sampling distribution
- c.) Probability Mass Function
- d.) Probability Density Function

### Sampling Distributions, Means, and Variances

(1) A random variable  $X$  has the following probability distribution

| X | P(X) |
|---|------|
| 1 | 0.25 |
| 2 | 0.25 |
| 3 | 0.25 |
| 4 | 0.25 |

- Derive the sampling distribution for the mean of  $n = 2$  observations of  $X$
- Compute the mean  $\mu_{\bar{X}}$  and standard deviation  $\sigma_{\bar{X}}$  of the sampling distribution of  $\bar{X}$

(2) A random variable  $Y$  has two possible outcomes. The outcome  $Y = 1$  occurs with probability  $p = 0.23$ . Derive the sampling distribution for the proportion of outcomes with  $Y = 1$  in a sample of  $n = 3$  observations of  $Y$ .

(3) Consider a survey of tree volume. Suppose that in the population of trees 10% have a volume of 10 cubic feet, 20% have a volume of 20 cubic feet, 30% have a volume of 30 cubic feet, and 40% have a volume of 40 cubic feet. Let  $X$  denote the volume of a randomly selected tree, assuming that every tree has an equal chance of being selected. The probability distribution of  $X$  is given in the table below.

| X  | P(X) |
|----|------|
| 10 | 0.1  |
| 20 | 0.2  |
| 30 | 0.3  |
| 40 | 0.4  |

- On **Homework 2** you confirmed that  $X$  has a mean of  $\mu = 30$  cubic feet and a standard deviation of  $\sigma = 10$  cubic feet. Derive the sampling distribution for the mean tree volume in a sample of  $n = 2$  trees.

## Deriving Probabilities From Discrete Distributions

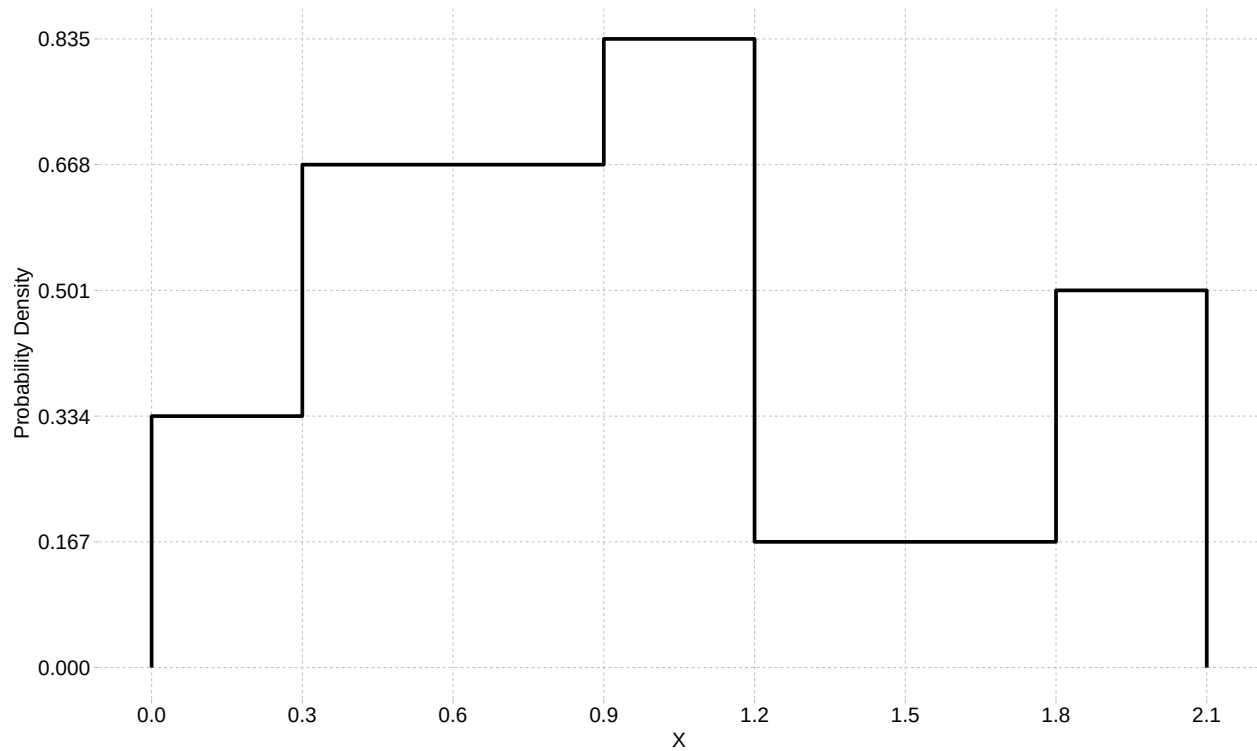
(4) Michael Dickson, who plays for the Seattle Seahawks, is a punter in the NFL. The following sampling distribution gives the mean punting distance (in yards) for a sample of  $n = 4$  punts. Use this distribution to answer the following questions:

| $\bar{X}$ = Mean Punt Distance | $P(\bar{X})$ |
|--------------------------------|--------------|
| 35.33                          | 0.40         |
| 45.67                          | 0.50         |
| 55.00                          | 0.08         |
| 60.33                          | 0.01         |
| 65.67                          | 0.01         |

- What is the probability the mean distance is 55 or more yards?
  - What is the probability the mean distance is at least 35.33 yards?
  - What is the probability the mean distance is between 35.33 and 45.67 yards?
  - What is the probability the mean distance is 35.33 yards or 55 yards?
- (5) A random variable  $X$  follows a binomial distribution with probability of success  $p = 0.43$  and number of trials  $n = 8$ . What is the probability of observing fewer than 3 successes, that is what is  $P(X < 3)$ ?
- (6) Tom Brady is a famous quarterback who recently retired from playing in the National Football League. During his career, the probability that Brady would complete a pass in a given play was 64.3%. If a typical NFL team runs about 35 offensive pass plays, what is the probability that Brady would complete exactly 28 pass plays in a game?
- (7) A random variable  $W$  follows a Poisson distribution with rate parameter  $\lambda = 1.8$  events per hour. What is the probability of observing exactly four events in a given hour, that is what is  $P(W = 4)$ ?
- (8) A factory experiences on average 3 equipment malfunctions per day, following a Poisson distribution with a rate parameter of 3. Calculate the probability that on a randomly selected day, the factory will observe 3 or more equipment malfunctions.

## Deriving Probabilities From Continuous Distributions

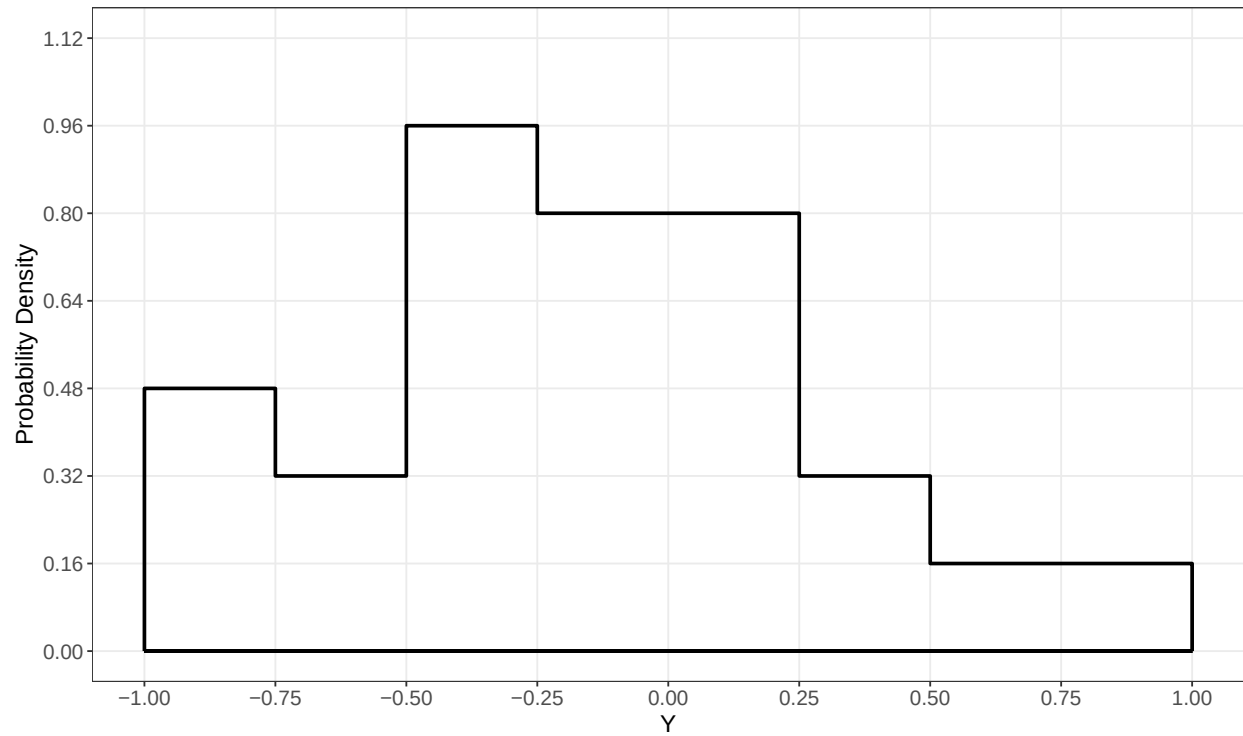
(9) Use the plot below of the probability distribution of a continuous random variable  $X$  to answer the following questions



Use the plot to find the following probabilities:

- What is  $P(X < 0.6)$ ?
- What is  $P(X > 0.3)$ ?
- What is  $P(1.5 < X < 1.8)$ ?

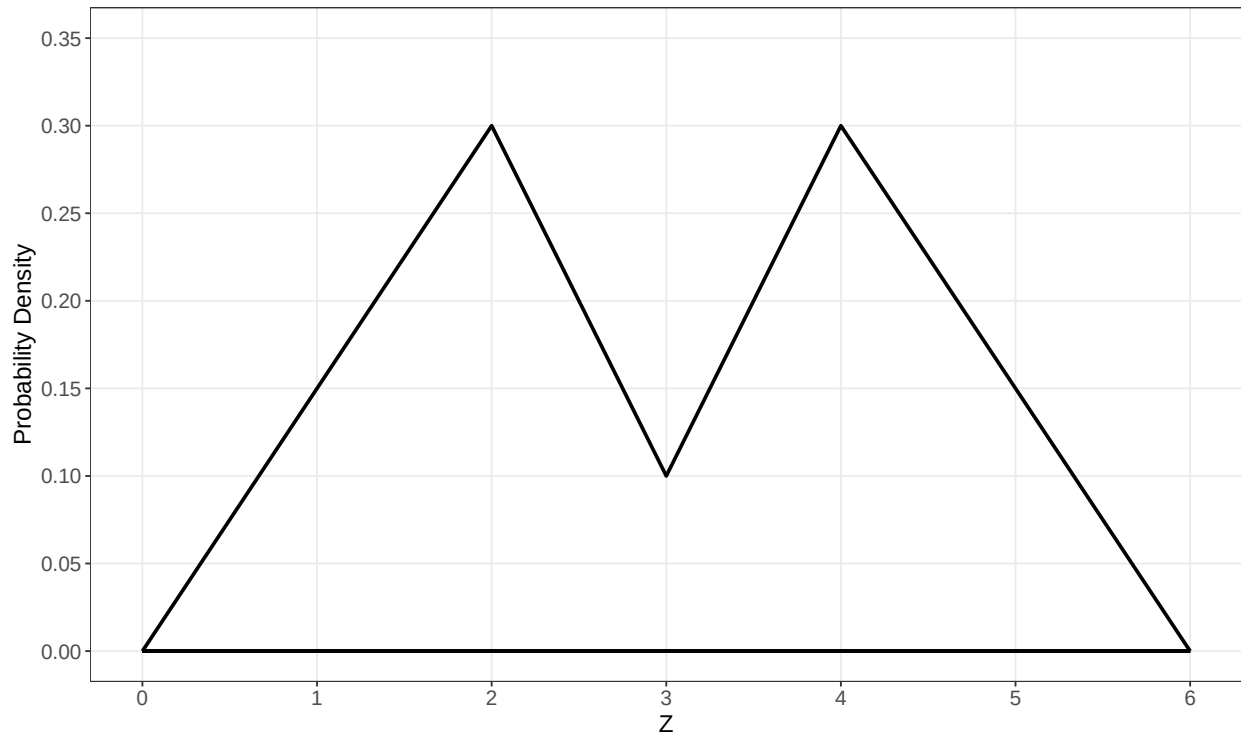
(10) Use the plot below of the probability distribution of a continuous random variable  $Y$  to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is  $\frac{1}{2} \times \text{base} \times \text{height}$ )

- What is  $P(Y > -0.25)$ ?
- What is  $P(-0.5 < Y < 0.25)$ ?

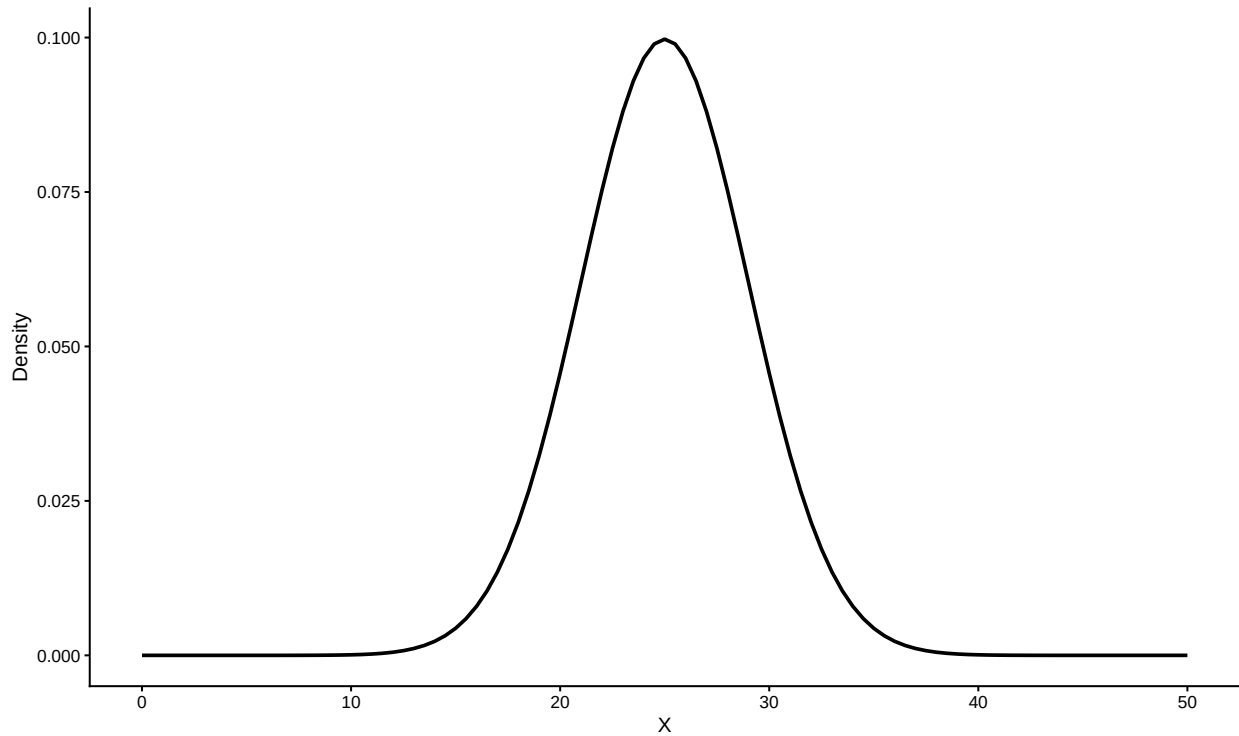
(11) Use the plot below of the probability distribution of a continuous random variable  $Z$  to answer the following questions



Use the plot to find the following probabilities (recall that the area of a triangle is  $\frac{1}{2} \times \text{base} \times \text{height}$ )

- What is  $P(Z < 2)$ ?
- What is  $P(2 < Z < 3)$ ?
- What is  $P(Z > 4)$ ?
- Give the approximate value of the 25th percentile of distribution of  $Z$

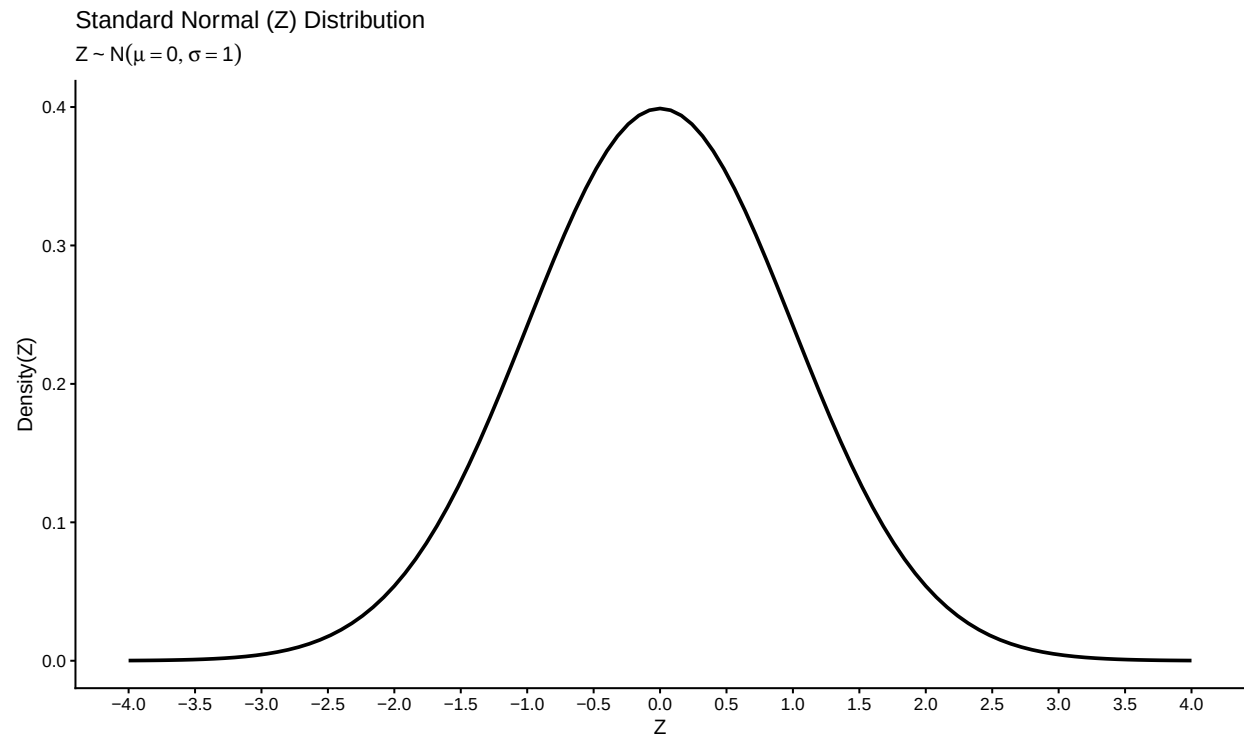
**12.)** Assume the continuous random variable  $X$  follows a normal distribution with mean  $\mu = 25$  and variance  $\sigma^2 = 16$ .



Using a  $Z$ -table, find the following (approximate) probabilities for  $X$ . You can also use the app in the course website or StatDistributions.com to check your result

- What is  $P(X < 15)$ ?
- What is  $P(X > 28)$ ?
- What is  $P(12 < X < 20)$ ?
- What is  $P(X > 30)$ ?
- What is  $P(25 < X < 33)$ ?
- What is the value of  $X$  that marks the 95th percentile of the distribution?
- What is the value of  $X$  that marks the 40th percentile of the distribution?

**13.)** Suppose that a random variable  $X$  follows a normal distribution with mean  $\mu = -5$  and standard deviation  $\sigma = 1.8$ . Using the plot of the  $Z$ -distribution below, shade in the area corresponding to the probability  $P(-2 < X < 0.5)$



## References